Chapter 20

Financial Globalization: Opportunity and Crisis

■ Chapter Organization

The International Capital Market and the Gains from Trade.

Three Types of Gains from Trade.

Risk Aversion.

Portfolio Diversification as a Motive for International Asset Trade.

The Menu of International Assets: Debt versus Equity.

International Banking and the International Capital Market.

The Structure of the International Capital Market.

Offshore Banking and Offshore Currency Trading.

The Shadow Banking System.

Banking and Financial Fragility.

The Problem of Bank Failure.

Government Safeguards against Financial Instability.

Moral Hazard and the Problem of "Too Big to Fail."

Box: The Simple Algebra of Moral Hazard.

The Challenge of Regulating International Banking.

The Financial Trilemma.

International Regulatory Cooperation through 2007.

Case Study: The Global Financial Crisis of 2007–2009.

Box: Foreign Exchange Instability and Central Bank Swap Lines.

International Regulatory Initiatives after the Global Financial Crisis.

Metrics for International Capital Market Performance

The Extent of International Portfolio Equity Diversification.

The Extent of Intertemporal Trade.

The Efficiency of International Asset-Price Arbitrage

The Efficiency of the Foreign Exchange Market.

Summary

■ Chapter Overview

The international capital market, involving Eurocurrencies, offshore bond and equity trading, and International Banking Facilities, initially may strike students as one of the more arcane areas covered in this course. Much of the apparent mystery is dispelled in this chapter. The chapter demonstrates that issues in this area are directly related to other issues already discussed in the course, including macroeconomic stability, the role of government intervention, and the gains from trade.

Using the same logic that we applied to show the gains from trade in goods or the gains from intertemporal trade, we can see how the international exchanges of assets with different risk characteristics can make both parties to a transaction better off. International portfolio diversification allows people to reduce the variability of their wealth. When people are risk-averse, this diversification improves welfare. An important function of the international capital market is to facilitate such welfare-enhancing exchanges of both debt instruments (such as bonds) and equity instruments (such as stocks).

Offshore banking activity is at the center of the international capital market. Central to offshore banking are Eurocurrencies (not to be confused with euros), which are bank deposits in one country that are denominated in terms of another country's currency. Relatively lax regulation of Eurocurrency deposits compared with onshore deposits allows banks to pay relatively high returns on Eurocurrency deposits. This has fostered the rapid growth of offshore banking. Growth has also been spurred, however, by political factors, such as the reluctance of Arab OPEC members to place surplus funds in American banks after the first oil shock for fear of confiscation by the U.S. government following the confiscation of Iranian deposits in 1979.

The text also introduces issues of regulating capital markets. Central to this task is the notion of how banks fail, and what can be done to prevent bank failures. Bank regulation presents a trade-off between financial stability and moral hazard. You want to promote confidence in the banking system through financial support, but too much support encourages risk taking by the banks. Deposit insurance, regulations, and lenders of last resort can all help prevent the lack of confidence in a banking system that can generate a run on the banking assets. International banking presents additional challenges as rules are not uniform, responsibility can be unclear, and enforcement is difficult. This tension is highlighted by the "financial trilemma," the observation that you can only ever have two of the following three policy goals: financial stability, national control over financial safeguard policy (e.g., FDIC insurance), and free capital mobility. If for example, one country was perceived to be more likely to bail out their national banking system, then this would trigger a flow of capital to that country as well as increase the risk-taking behavior of that nation's banks, reducing financial stability.

Industrialized countries are involved in an effort to coordinate their bank supervision practices to enhance the stability of the global financial system. Common supervisory standards set by the Basel Committee were developed. Potential problems remain, however, especially regarding the clarification of the division of lender-of-last-resort responsibilities among countries and the increasingly large role of nonbank financial firms, which makes it harder for regulators to oversee global financial flows. The text highlights these regulatory difficulties using a case study of the Global Financial Crisis of 2007-2009. This case study illustrates the difficult balance regulators face between creating moral hazard and maintaining financial stability. The financial crisis of 2007–2009 also highlights the increasingly important role played by nonbank financial institutions, the so-called "Shadow Banking System." Though these institutions operate much like banks and their profits are intertwined with those of commercial banks, they are not regulated like commercial banks. Much of the riskiest behavior that contributed to the financial crisis was initiated by these institutions. In response, the U.S. Congress passed the Dodd-Frank Act, which allows the government to regulate these institutions like banks. This represents another example of the difficulty in balancing financial support (the bailouts of financial institutions) with moral hazard (increased supervision and regulation).
The global aspect of the financial crisis is also highlighted with a discussion of Central Bank Swap Lines. European banks heavily invested in mortgage-backed securities because they were given good credit ratings and thus allowed these banks to hold less capital against their purchases of these assets. However, these

banks did not want exposure to currency risk, so they financed their purchases by borrowing dollars in short-term markets. When mortgage-backed securities plummeted in value, these European banks were faced with a dilemma. They could not be bailed out by their local central banks because they needed to pay back their debts in dollars. However, they did not want to sell their dollar-denominated assets at such a low price. To resolve this dilemma, the Federal Reserve stepped in and lent central banks around the world dollars, which they could in turn use to bail out their local commercial banks. This demonstrates an important aspect of increased capital mobility: the importance of policy coordination across countries.

The evidence on the functioning of the international capital market is mixed. International portfolio diversification appears to be limited in reality despite the apparent gains in terms of reduced risk from international diversification. Intertemporal trade also appears to be lacking, as evidenced by the strong correlation between national savings and investment rates. Fully integrated capital markets should allow countries' savings and investment rates to diverge as savings seeks out its most productive use worldwide, regardless of location.That said, the observed strong positive correlation between savings and investment rates has considerably weakened when examined using contemporary data. Another test of international capital market integration is whether onshore and offshore interest rates on similar assets denominated in the same currency differ considerably. Prior to the Global Financial Crisis of 2007-2009, these interest rates were very close and highly correlated. However, since 2009, we have observed significant deviations in these interest rates, likely as a result of financial regulations that limit international arbitrage between these assets.

The recent performance of one component of the international capital market, the foreign exchange market, has been the focus of public debate. Government intervention might be uncalled for if exchange-rate volatility reflects market fundamentals but may be justified if the international capital market is an inefficient, speculative market, drifting without the anchor of underlying fundamentals. The performance of the foreign exchange markets has been studied through tests of interest parity, tests based on forecast errors, attempts to model risk premiums, and tests for excess volatility. Research in this area presents mixed results that are difficult to interpret, and there is still much to be done.

■ Answers to Textbook Problems

1. The better diversified portfolio is the one that contains stock in the dental company and the dairy company. Good years for the candy company may be correlated with good years for the dental company, and conversely, the return from a portfolio consisting of these stocks would be more volatile than the return from a portfolio consisting of the dental and dairy company stocks.

2. Our two-country model (Chapter 19) showed that under a floating exchange rate, monetary expansion at home causes home output to rise but foreign output to fall. Thus, national outputs (and earnings of companies in the two countries) will tend to be negatively correlated under a floating rate if all shocks are monetary in nature. Chapter 18 suggests, however, that the correlation will be positive under a fixed rate if all shocks are monetary. The gains from international asset exchange are, therefore, likely to be greater under floating exchange rates, under the conditions assumed.
3. The main reason is political risk—as discussed in the Appendix to Chapter 14.
4. Reserve requirements are important for bank solvency. Maintaining adequate reserves enables a bank to remain solvent, even in periods in which it faces a relatively high amount of withdrawals relative to deposits. The higher the reserve requirements faced by a bank, however, the lower the bank's profitability. To create a "level playing field" for U.S. banks with foreign branches as compared to U.S. banks without foreign branches, it is important to ensure that banks with foreign branches cannot shift around their assets in a manner that reduces their reserve requirements, an option not open to U.S. banks without foreign branches.
5. This is again an open-ended question. The main criticism of Swoboda's thesis is that foreign central banks held dollars in interest-bearing form, so the United States extracted seigniorage from issuing

reserves only to the extent that the interest it paid was less than the rate it would have paid were the dollar not a reserve currency. The high liquidity of the dollar makes this plausible, but it is impossible to say whether the amount of seigniorage the United States extracted was economically significant.

6. Tighter regulation of U.S. banks increased their costs of operation and made them less competitive relative to banks that were not as tightly regulated. This made it harder for U.S. banks to compete with foreign banks and led to a decline in U.S. banking in those markets where there was direct, unregulated foreign competition.
7. Banks are more highly regulated and have more stringent reporting requirements than other financial institutions. Securitization increases the role played by nonbank financial institutions over which regulators have less control. Regulators also do less monitoring of nonbank financial institutions. As the role of nonbank financial institutions increases with securitization, the proportion of the financial market that bank regulators oversee declines, as does the ability of these regulators to keep track of risks to the financial system.
8. The extent of international diversification should go down because some consumption now depends exclusively on local conditions. In that case, agents want assets correlated with the price of that consumption. Local raspberry-oriented assets would rise in price when raspberries were expensive and fall when they were inexpensive. This helps consumers smooth their consumption of raspberries. They would still want some international diversification, but the presence of some nontraded consumption implies a lower level of diversification.
9. No, real interest rate equality is not an accurate barometer of international financial integration. As we saw in Chapter 16, there is a real interest parity condition, which is that $r = r^* + \%\Delta^e q$. Where r is the home real interest rate, r^* is the foreign, and $\%\Delta^e q$ is the expected percentage change in the real exchange rate. If there are real differences between countries' productivity or trends in world demand that lead to expected changes in the real exchange rate over time, we may see different real interest rates despite perfectly functioning integrated financial markets.

10. The majority of U.S. foreign debt is denominated in dollars, while a significant portion of U.S. foreign assets are in foreign currency. Thus, a depreciation of the dollar would not have much of an impact on the dollar value of U.S. foreign debt, since it is already denominated in dollars. However, the dollar value U.S. foreign assets would increase as the dollar depreciates since the foreign currency value of these assets are now worth more dollars. As such, the dollar value of assets increases more than the dollar value of debt and thus net foreign debt will increase less than the cumulative value of current account deficits.

11. U.S. gross foreign liabilities rise as the Brazilian has a claim on the fund, and U.S. assets rise as the fund buys more Brazilian equity. Likewise, Brazil's foreign assets and liabilities rise. But no U.S. citizen has altered his foreign holdings, and the Brazilian has bought Brazilian stocks. It is true that the market is more globalized, but individual agents are no more diversified.
12. This problem presents a trade-off between a bank's desire to put as much of its operating capital to work earning a return and its desire to signal strong financial solvency. There are higher potential profits to be made when the bank finances its purchase of risky assets through borrowing. However, doing so reduces the bank's ratio of capital to assets, lowering the threshold losses at which the bank becomes insolvent. Given the greater risk of this strategy, the bank will have to compensate its depositors with higher interest rates. These rates may be enough to actually make financing the asset purchase through a stock issue the better option.
13. The scenario in the previous problem changes if the bank's creditors expect the government to bail out the bank if it becomes insolvent. In this case, the bank would in fact be better off financing its operations through additional borrowing because the interest it would pay on borrowing would now be lower. If the bank's creditors are confident that the government will make good on the bank's losses in the event of insolvency, then they do not need to be compensated nearly as much for the bank's risky behavior.

14. Eurodollar interest rates exceed those on U.S. bank deposits after the global financial crisis because investors perceive greater risk in Eurodollar deposits than in U.S. deposits. The perception of greater risk can be related to the perception that the U.S. government had a greater willingness and likelihood to bail out struggling American banks than their counterparts in Europe do.